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Paleomagnetic Study of some Cretaceous and Tertiary Rocks on Hispaniola

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Summary – A reconnaissance paleomagnetic study of Hispaniola shows that three igneous units in the Dominican Republic possess meaningful directions of magnetism. A Late Cretaceous tonalite, an Eocene pyroxene diorite and a Miocene andesite porphyry have been investigated. The rock material studied is fresh, and has not been affected by secondary oxidation except in the case of the andesite which is occasionally weathered and reveals some hydrothermal alteration. Alternating field and thermal demagnetization result in removal of viscous remanence in some samples, while others reveal a good stability of NRM and little change in direction. The results disclose directions of magnetization substantially different from that of the present earth's field in Hispaniola and from those obtained from contemporaneous rocks of North America. They yield paleomagnetic poles at 23.1° N, 144.9° W for the Cretaceous tonalite and at 17.4° N, 138.0° W for the Eocene diorite, the positions of which are not significantly different from each other, suggesting no change of geomagnetic field direction during the two epochs. These poles have generally similar positions to those obtained from Late Cretaceous rocks on Jamaica and Puerto Rico. The Miocene data fall into two groups, one having a direction corresponding to a pole closely coinciding with the Miocene North American pole and the other giving a paleomagnetic pole at 68.3° N, 151.9° W coinciding with the Miocene pole for Jamaica. Paleotectonic interpretation of the results suggests that like other Greater Antilles, Hispaniola has been subject to large anticlockwise rotation since Late Cretaceous.

Key words: Paleomagnetism of Hispaniola.

1. Introduction

As an extension of the paleomagnetic investigations of Jamaica (VINCENZ *et al.*, 1973; STEINHAUSER and VINCENZ, 1973; DASGUPTA and VINCENZ, 1975; GUJA and VINCENZ, 1977; VINCENZ and DASGUPTA, 1977), a reconnaissance paleomagnetic study of Hispaniola was made. Hispaniola is one of the largest islands of the Greater Antilles and by virtue of its geographic position holds the key to the proper understanding of the tectonics of the Caribbean 'plate'. The objective of the study was to ascertain the suitability of rocks on Hispaniola for paleomagnetic purposes and to obtain more paleomagnetic data for the Caribbean in order to evaluate its plate tectonic character.

Paleomagnetic studies of Jamaica have suggested that Jamaica has been subject to both translational and rotational motions since the Cretaceous. These motions appear

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to be related to the motion of the Caribbean 'plate' as a whole, though they appear mainly as semi-independent rotations of Jamaica as a marginal block of the 'plate'. Such rotations are implied by the equatorial directions of magnetization found in some Cretaceous rock units on Jamaica and by analogous results obtained in Puerto Rico (FINK and HARRISON, 1971) and in southern and western Caribbean (HARGRAVES and SHAGAM, 1969; HARGRAVES and SKERLEC, 1976; MACDONALD and OPDYKE, 1972; GOSE and SWARTZ, 1977). Rotational movements were also deduced from paleomagnetic data in the Mediterranean (VAN DER VOO, 1969; SOFFEL, 1972; DE JONG *et al.*, 1973; VAN DEN BERG *et al.*, 1975), and the question arises whether this effect is not typical for plate margins when two plates are not in a direct collision, but are sliding in opposite directions along their line of contact.

Since Hispaniola forms a separate tectonic block on the margin of the Caribbean, it is worthwhile to test this hypothesis at least for the case of a 'miniplate' such as the Caribbean.

2. *Geology*

As in Jamaica the geology of Hispaniola is complex and not very well known. BUTTERLIN (1956) and WEYL (1966) have described its general aspects and BOWIN (1966) has mapped in some detail Late Cretaceous and Eocene rocks in Central Dominican Republic. More recently MACDONALD and MELSON (1969) described the Miocene intrusives and Recent lavas in Eastern Dominican Republic.

In a paleomagnetic sampling reconnaissance conducted in 1972 a number of intrusives and lavas were sampled in the Dominican Republic and Haiti. Three units in the Dominican Republic were found to possess well exposed fresh outcrops and were selected for more detailed study. These comprise the Late Cretaceous El Rio tonalite pluton, the Eocene pyroxene diorite intrusion of LaJina and the Late Miocene andesite prophyry intrusive of Las Yayas de Viajama (Fig. 1). According to available geological information (BOWIN, 1966: MACDONALD and MELSON, 1969; PALMER, 1972; LEWIS, 1977), none of these units appear to have been structurally deformed and the first two are quite fresh, while the andesite is somewhat weathered in exposures in elevated positions, but not in stream beds. The tonalite pluton is of the unfoliated type and could have only been affected by a dip slip movement associated with the Bonao fault situated to the east of it, but without tilting or rotation. Similarly, the diorite intruded along the Hatillo thrust fault is younger than the faulting and thus unaffected by any movements. No tectonic movements of any significance appear to have affected the Miocene andesite.

Opaque mineralogy of the three units was studied under a reflected light microscope. Despite its coarse texture, the El Rio tonalite contains non-uniformly and sparsely disseminated titanomagnetite with grain diameters ranging from a few to about 30 μm with the former predominating. Study of larger grains disclosed class 2



Figure 1
Sketch map of Hispaniola. E, Late Cretaceous El Rio Tonalite; D, Eocene LaJina pyroxene diorite; Y, Miocene andesite of Las Yayas de Viajama.

deuteric oxidation index of the titanomagnetite. The deuteric oxidation class is related to the degree of high temperature oxidation of titanomagnetite grains as observed quantitatively under the microscope in polished sections. According to this classification class 1 grains consist of homogeneous unoxidized titanomagnetite, while higher classes 2 to 6 reveal progressively increasing number of ilmenite lamellae, which eventually become altered, and progressive alteration of titanomagnetite to titanohematite or hematite and pseudobrookite as the final product (WILSON and WATKINS, 1967; WILSON *et al.*, 1968; ADE-HALL *et al.*, 1968). None of the grains showed any hydrothermal alteration (ADE-HALL *et al.*, 1971). The titanomagnetite grains in polished sections of the Eocene diorite are smaller (5 to 10 μm), but their oxidation characteristics are similar to those of the tonalite. Both intrusions contain non-magnetic or weakly magnetic grains of a bright yellow sulphide mineral. The Yayas andesite contains more densely and uniformly disseminated class 2 and 3 fine grained (diameters 5 to 50 μm) titanomagnetite with an overprint of low temperature oxidation indicated by granulation which is ascribed to mild hydrothermal alteration (ADE-HALL *et al.*, 1971). All three units reveal absence of significant amounts of maghemite so typical for the rock units on Jamaica where they have been ascribed to weathering (DASGUPTA and VINCENZ, 1975; GUJA and VINCENZ, 1977).

3. Sampling and laboratory measurements

To ensure access to fresh rock exposures, whenever possible sampling was conducted in outcrops situated in stream beds or on river banks.

The El Rio tonalite was sampled in five sites designated EA, EB, EC, ED and EE (Table 1) and situated in Rio Jimenoa over a distance of about 3 km northeast of El

Table 1

Directions of magnetization

Unit	Site	Virgin samples					Demagnetized samples				
		N	D	I	α_{95}	k	N	D	I	α_{95}	k
	EA	3	290.8	52.5	49.6	7.2	3	287.9	34.6	24.3	26.7
Late	EB	19	323.2	49.2	8.3	17.2	13	294.8	46.7	6.3	43.9
Cretaceous	EC	17	286.7	36.0	10.3	13.0	16	282.3	36.3	5.7	42.4
tonalite	ED	21	333.6	62.0	11.8	8.2	6	290.1	30.3	14.5	22.1
	Mean	4	305.6	51.6	20.8	20.4	4	288.6	37.1	9.1	101.9
	DA	13	285.8	41.1	39.8	2.0	6	282.8	43.4	26.7	7.3
Eocene	DB	11	313.2	50.4	24.9	4.3	6	285.2	47.0	14.3	22.8
diorite	DC	11	286.0	60.4	31.7	3.0	6	275.3	40.7	25.1	8.1
	Mean	3	295.0	51.3	21.4	34.3	3	280.1	43.7	7.5	272.3
	YA	11	154.6	−48.1	31.6	3.0	10	184.9	−43.8	20.7	6.4
	YB	11	175.9	−50.5	10.5	19.8	10	179.2	−46.5	5.4	80.7
Miocene	YC	16	145.3	−29.8	20.4	39.8	15	154.3	−35.4	6.0	40.9
andesite	YD	10	161.5	−32.8	8.4	33.6	9	161.9	−36.9	5.2	98.3
	Mean	4	158.3	−40.8	16.3	32.6	4	169.3	−41.3	11.5	44.8
	YA + YB	2	—	—	—	—	2	182.1	−45.2	(10.6)	(559.5)
	YC + YD	2	—	—	—	—	2	158.0	−36.2	(13.8)	(380.4)

Note: N is the number of samples or sites; D the declination, I, the inclination and α_{95}, the 95 percent radius of confidence all given in degrees of arc; k is the estimate of the precision parameter (FISHER, 1953).

Rio. The natural remanence (NRM) of a total of 60 samples was measured, but the intensities of some samples became too weak after alternating field (AF) or thermal demagnetization and could not be measured on the equipment available. These samples had to be rejected as well as a few which had been remagnetized in the general direction of the present earth's field. This eliminated six samples from site EB and one sample from site EC. Fifteen samples from site ED were derived from an outcrop of altered tuffaceous late Cretaceous Tireo formation (BOWIN, 1966) where it has been invaded by the tonalite. These gave inconsistent directions of magnetization and were also rejected. Eight tonalite samples from site EE (not shown in Table 1), though fairly strongly magnetized, all carried magnetization of recent origin, which could not be removed by AF or thermal treatment. The reason for their apparent remagnetization is unknown and they were also discarded, leaving a net total of 38 samples for final analysis.

The LaJina diorite was sampled on the banks of Rio Ozama between LaJina and Serrales. Five sites designated DA, DB, DC, DD and DE were occupied, but again some samples, particularly all those from sites DD and DE, were either unstable or very weakly magnetic so that the results for only three sites could be used for a net collection of eighteen samples.

The Yayas andesite was sampled in five sites east and northeast of Las Yayas de

Viajama designated YA, YB, YC, YD and YE, but YE located in weathered rock on the hill side gave scattered directions of magnetization. As a result the data for only four sites were used in this investigation, two on the banks of Arroyo Viajama and two on the hillside, comprising a net total of 44 samples.

4. Experimental results

Measurements were conducted using a Princeton Applied Research, Inc. (PAR) Model SM-1 spinner magnetometer. The maximum error of measurement was $\pm 10^{-3}$ A/m in SI units (1 A/m = 10^{-3} e.m.c.g.s. unit of magnetization) giving an accuracy of better than $\pm 2°$ in direction for intensities greater than 100×10^{-3} A/m. AF demagnetization was conducted using apparatus with a three-axis tumbler, capable of developing peak fields up to 5000 Oe (1 Oe = $10^3/4\pi$ A/m). Thermal demagnetization was carried out using the equipment previously described by VINCENZ *et al.* (1965). The residual field within the field free space was variable and non-uniform with maximum amplitude of ± 50 gammas.

The tonalite is weakly magnetized, its NRM averaging $131.7 \pm 156.7 \times 10^{-3}$ A/m. The intensities of the diorite average $52.2 \pm 46.9 \times 10^{-3}$ A/m. The Miocene andesite is on the other hand fairly magnetic, averaging in intensity $238.4 \pm 131.2 \times 10^{-3}$ A/m.

Due to small magnetic grain sizes, the magnetization of the tonalite and the diorite is fairly stable and in the majority of samples does not change much in direction on AF or heat treatment. Some samples did, however, carry viscous remanence (VRM) and stable directions of measurable intensities were achieved using peak fields of 88, 175 or 350 Oe or temperatures of 100°, 320° or 420°C. In many cases the directions stabilized when the intensities were reduced to about 10×10^{-3} A/m, but more often stable directions were achieved using peak fields of 88 Oe or temperatures of 320°C. The more magnetic andesite also revealed a high directional stability in some samples, while in others directions stabilized on treatment with peak fields up to 350 Oe or on heating to 420°C.

Figure 2 shows typical AF and thermal demagnetization spectra and Table 1 summarizes directional changes before and after demagnetization. The table gives also a comparison of the mean directions for each unit before and after demagnetization, including stable virgin directions which did not change on AF or heat treatment. It should be noted that some spectra in Fig. 2 appear to be incomplete, but in reality reflect the limitations imposed by the limits of sensitivity of the PAR spinner used. In the tonalite, the more strongly magnetized samples carried secondary NRM only and weak intensities ($20–50 \times 10^{-3}$ A/m) were primarily those corresponding to the original cooling thermoremanence (TRM). Thus at temperatures of 500°C or peak demagnetizing fields of 700 Oe, and in some cases even 175 Oe, the primary intensities of the tonalite and diorite samples became too weak to be measured

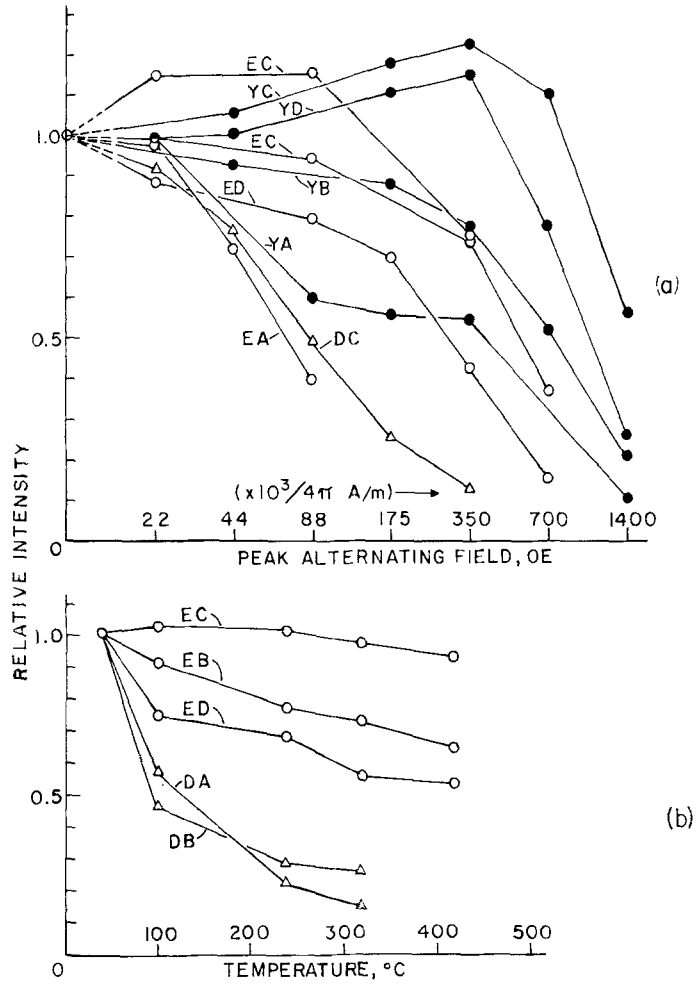


Figure 2
Demagnetization spectra. (a) AF demagnetization, (b) thermal demagnetization; tonalite: EA, EB, EC and ED; diorite: DA, DB and DC; andesite: YA, YB, YC and YD.

accurately. Fortunately, in most cases the directions stabilized after demagnetizing with peak fields of 350 Oe or after heating to 320°C and hence this shortcoming did not significantly interfere with achieving stable directions of magnetization. The more strongly magnetic andesite samples were demagnetized using peak fields up to 1400 Oe, but stabilization of direction was achieved usually at 175 Oe or a maximum of 350 Oe. Some directions did not change on demagnetization, but those that did revealed a distinct movement away from the present-day geomagnetic field direction. This corresponded to removal of a VRM or a CRM (chemical RM) from this reversely magnetized rock, resulting in an increase of NRM as shown by the curves for YC and YD in Fig. 2.

Table 1 shows that except for some decrease in scatter, there is little difference in the mean directions before and after magnetic or thermal 'cleaning'. Any changes can be accounted for by removal of VRM causing the directions to swing away from the present-day field direction.

5. *Discussion*

The results in Fig. 2 and Table 1 show that the primary NRM of the tonalite, though weak, is quite stable and has a different direction from that of the present-day geomagnetic field in Hispaniola (compare with Fig. 3). While at one site and in some samples the tonalite appears to have been remagnetized, in most cases its secondary magnetization is not appreciable and is probably a VRM. The fairly large directional

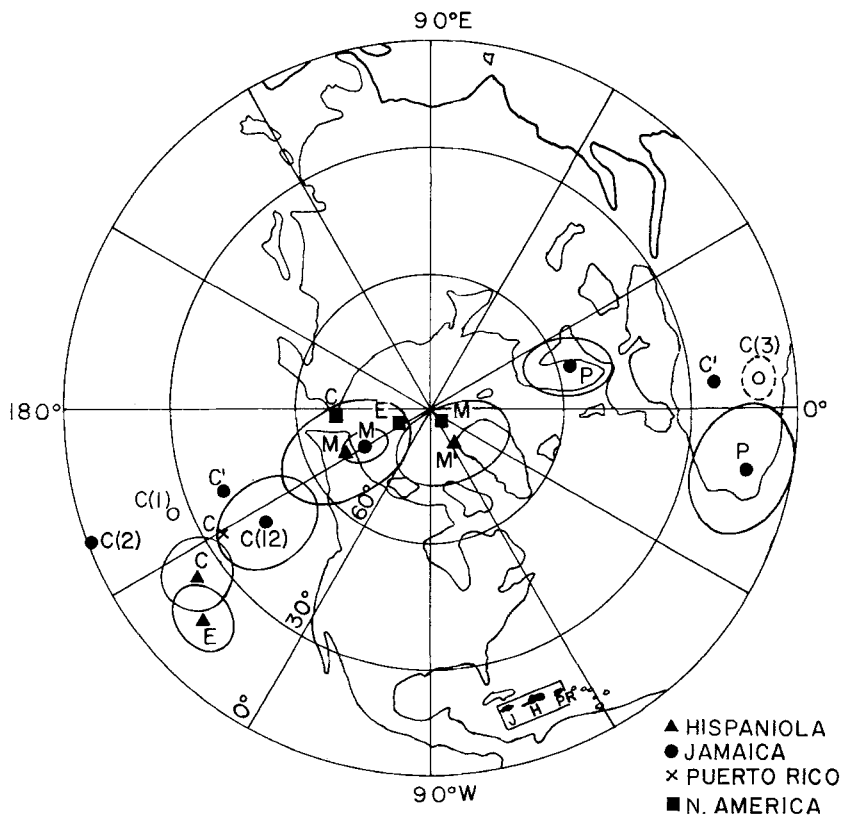


Figure 3

Paleomagnetic poles: C, Cretaceous, P, Paleocene, E, Eocene, M, Miocene; C(12), Cretaceous pole for twelve units on Jamaica; C′, C(1), C(2) and C(3): low latitude poles for Jamaica; C″, an intermediate latitude pole for Jamaica; M′, Miocene pole for Hispaniola obtained from sites YA and YB, while the other Miocene pole has been obtained from sites YC and YD.

scatter even after demagnetization must be ascribed to experimental imperfections related to the weak intensities measured after AF or thermal 'cleaning', since for intensities of the order of 10×10^{-3} A/m the expected overall directional error would be $\pm 5°$.

The magnetization of the diorite is less stable, but the directional scatter is similar to that of the tonalite. It is of interest to note that a statistical test shows that there is no significant difference between the mean directions of the two rock units. The two values of the precision parameter k obtained after demagnetization can be compared using the F test (WATSON, 1956) and an F of 2.68 is obtained which is less than $F_{6,4} = 6.26$ at the 5 percent probability level. This lack of significant difference between the two k values permits application of the F test to the two mean directions which are also not significantly different at the 5 percent probability level, yielding $F = 3.4$ which is less than $F_{2,10} = 4.0$ (WATSON, 1956). The test suggests that there was no significant difference between the directions of geomagnetic field in Hispaniola at the time of the injection of the two intrusions. Further statistical tests show, however, that these directions differ significantly from the mean Cretaceous direction obtained from twelve rock units on Jamaica (VINCENZ *et al.*, 1973). While no definite K-Ar dates are available for either of the two units, (a date of 86 m.y. has been quoted in a private communication from S. E. Kessler) field geological evidence gives the ages of the tonalite and the diorite as Late Cretaceous and Mid to Late Eocene respectively. Thus the paleomagnetic data, though based on only two units, suggest that the direction of geomagnetic field in Hispaniola did not change significantly during the time interval between Late Cretaceous and at least Mid-Eocene. This implies that there was no significant movement of Hispaniola during this time interval.

The results for the Late Miocene andesite are ambiguous. Table 1 shows that AF treatment has separated the directions of magnetization into two distinct groups and it is not clear which of the two directions is primary. The mean for YA and YB corresponds to an approximate reversal of the present-day field, while that for YC and YD agrees closely with the direction obtained from Miocene lavas on Jamacia (GUJA and VINCENZ, 1977), see also Fig. 3. It is conceivable that the mild regional hydrothermal alteration of the andesite may have been late post-formational and could have affected the original cooling TRM. The alteration is indicated by granulation of titanomagnetite without affecting the ilmenite lamellae and it is usually accepted that this type of change is associated with reheating to a maximum of about 200°C (ADE-HALL *et al.*, 1971). This could have imposed a partial TRM or a CRM of more recent origin. The stability of this secondary remanence resulting from an oxidation process may be higher than that of the primary remanence and AF treatment might be ineffective in removing it. In such a case the 'clean' NRM would still differ in direction from the original cooling TRM. It is, however, not clear which of the two directions should be ascribed to this effect. A very recent reversed field could have imposed the direction represented by the mean direction of YA and YB sites (Table 1). Alternatively, AF treatment did not quite remove the effect of the present-day field from the YC and YD

samples. Since the latter mean direction almost coincides with that found for Late Miocene lavas on Jamaica, we tend to think that it does represent the Late Miocene field in Hispaniola. In this case the direction represented by the mean of YA and YB reflects the effect of partial remagnetization. Unfortunately, statistical significance cannot be assigned to the two mean directions based on data from two groups of two sites and more data are required to clarify this issue.

6. *Paleotectonic considerations*

The results are summarized in Table 2 and are compared with paleomagnetic results from other Greater Antilles in Fig. 3. They show that rocks on Hispaniola give

Table 2
Paleomagnetic results

Rock unit	N	D	I	α_{95}	k	λ'	φ'	δp	δm	Polarity
Cretaceous tonalite	4	288.6	37.1	9.1	101.9	144.9° W	23.1° N	6.5	10.7	−
Eocene diorite	3	280.9	43.7	7.5	272.3	138.0° W	17.4° N	6.2	9.5	−
Miocene andesite										
(YA + YB)	2	182.1	−45.2	(10.6)	(559.5)	58.4° W	81.3° N	(9.0)	(13.6)	+
(YC + YD)	2	158.0	−36.2	(13.8)	(330.4)	151.9° W	68.3° N	(9.8)	(16.1)	+

Note: N is the number of sites, λ' and φ' the longitude and latitude of the paleomagnetic pole and δp and δm the semi-axes of its confidence oval in degrees of arc; − and + give normal and reversed polarity of the pole respectively; other symbols as in Table 1.

meaningful paleomagnetic information, suggesting that further studies are worthwhile and should be continued. In Fig. 3 the paleomagnetic poles for Hispaniola are shown by triangles the positions of which are compared with the poles available from North America, Jamaica and Puerto Rico. The figure shows that the Cretaceous and Eocene poles for Hispaniola differ from the Cretaceous poles for Jamaica designated C(12), with the Puerto Rican pole (FINK and HARRISON, 1971) falling in between. Some of the Jamaican Cretaceous poles have, however, low latitude positions similar to those for Hispaniola (C′, C(1), C(3)) as well as intermediate latitude positions (C″). All these poles lie on a great circle which passes through the Paleocene (DASGUPTA and VINCENZ, 1975) and Miocene (GUJA and VINCENZ, 1977) Jamaican poles as well as through the North American poles of corresponding ages (HICKEN *et al.*, 1972). The figure shows that all poles for Hispaniola have positions significantly different from the contemporaneous poles for North America. An exception to this is pole M' obtained from the outcrops at sites YA and YB, which might represent a late remagnetization, and almost coincides with the Miocene North American pole. All these data suggest that Hispaniola, Jamaica and Puerto Rico have been subjected to large anticlockwise rotations since the Cretaceous relative to North America. The rotations can

be viewed in the light of the interpretation suggested previously for the Jamaican data (GUJA and VINCENZ, 1977), according to which part of the rotation could be ascribed to the translational motion of the Caribbean 'plate' from the southwest and a substantial part to the semi-independent rotations of the individual blocks represented by the islands. There is at present no unique way of discriminating between the proposed motions, i.e. it is not possible to say how much of the rotation (if any) should be ascribed to the rotation of the whole Caribbean as a result of its motion relative to North America (Eulerian motion) and how much has been due to local rotations alone. Since rotations have also been deduced from paleomagnetic data for the southern and western margins of the Caribbean (MACDONALD and OPDYKE, 1972; HARGRAVES and SKERLEC, 1976; GOSE and SWARTZ, 1977), both effects may be assumed to have taken place. It is, however, not possible to conclude from the present data that rotations at plate margins are common features of all tectonic plates, though the present data strongly suggest that they have occurred in the Caribbean. The data indicate also that there has been no significant movement of Hispaniola between Late Cretaceous and Mid-Eocene. Recent discovery of the spreading centre in the Cayman Trough (HOLCOMBE *et al.*, 1973; HEIRTZLER, 1976; BALLARD, 1976), and seismic evidence (MOLNAR and SYKES, 1969) have confirmed the plate tectonic character of the Caribbean and paleomagnetic data for the Greater Antilles fit this interpretation. The results presented here will, hopefully, lead to more detailed paleomagnetic studies of Hispaniola and of the Caribbean as a whole.

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